Project Proposal

Reducing Hallucination in Legal RAG Chatbots:

A Comparative Study of Deep Learning Retrieval Architectures

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Data

We plan to use two primary U.S.-based public legal datasets.

(1) Harvard Caselaw Access Project (CAP) (case.law/caselaw), which provides 6.7 million digitized U.S. court decisions spanning 360 years under a CC0 license, available as bulk JSON via the Caselaw Access Project API and on HuggingFace (HuggingFace: free-law/Caselaw_Access_Project). A pre-built FAISS vector index is also publicly available.

(2) Pile of Law (HuggingFace: pile-of-law/pile-of-law), a 256GB curated corpus containing CourtListener court opinions, r/legaladvice Q&A pairs, U.S. federal register filings, USPTO office actions, and HHS administrative law judge opinions. For retrieval evaluation, we will use **LePaRD** (Legal Passage Retrieval Dataset), which contains millions of U.S. federal judge citations with ground-truth passage-level retrieval pairs.

Background and Motivation

Large Language Models (LLMs) integrated with Retrieval-Augmented Generation (RAG) have shown promise for legal information retrieval, yet hallucination remains a critical barrier to adoption in high-stakes legal contexts. The 2023 *Mata v. Avianca Airlines* case, where an attorney submitted a brief citing non-existent cases generated by ChatGPT, underscored the real-world consequences of hallucinated legal content. While RAG mitigates hallucination by grounding generation in retrieved documents, the quality of the retrieval step itself varies significantly depending on the underlying deep learning architecture used for encoding and matching. This project is motivated by the need to systematically compare how different neural architectures perform at the retrieval stage of a legal RAG pipeline, directly measuring their impact on hallucination rates.

Problem

Which deep learning retrieval architecture most effectively reduces hallucination in a legal advice RAG chatbot? Specifically, we compare retrieval quality and downstream generation accuracy across:

(a) TF-IDF/Word2Vec baselines,

(b) CNN-based text encoders,

(c) RNN/LSTM sequence encoders,

(d) Transformer-based bi-encoders (BERT), and

(e) Knowledge Graph-augmented retrieval.

We measure whether improved retrieval precision translates into measurably lower hallucination rates in generated legal responses.

Scope and Methods

The project implements a modular RAG pipeline where the retrieval component is swappable across five architecture classes.

(a) **Baseline:** TF-IDF sparse retrieval and Word2Vec/GloVe dense retrieval using cosine similarity over the CAP corpus.

(b) **CNN Retriever:** 1D convolutional text encoder with max-pooling to produce fixed-size document embeddings, trained on legal passage pairs.

(c) **RNN/LSTM Retriever:** Bidirectional LSTM encoder capturing sequential dependencies in legal text, with attention-weighted hidden states as document representations.

(d) Transformer

(e) **Retriever:** Fine-tuned BERT bi-encoder (or Legal-BERT) producing contextual embeddings stored in a FAISS vector database for semantic search.

(f) **Knowledge Graph-Augmented:** Entity extraction and relation triples (subject-verb-object) from legal texts to build a citation knowledge graph, combined with Transformer retrieval for hybrid grounding.

Each retriever feeds into the same LLM generator (e.g., Llama 2 or Mistral via HuggingFace) to isolate the retrieval variable. Evaluation uses LePaRD ground-truth citations to measure Recall@k and Precision@k at the retrieval level, and human-annotated hallucination rate, faithfulness score, and F1 on generated answers. The chatbot frontend will be deployed via Streamlit. Implementation uses PyTorch, HuggingFace Transformers, LangChain, FAISS, spaCy, and Gensim.

Concerns & Limitations

1. **Ethical:** All datasets are publicly available and do not require NDAs. The chatbot will include disclaimers that it does not constitute legal advice. Court opinions may contain personally identifiable information; we will follow the data providers' existing redaction practices.
2. **Computational:** The full CAP corpus (6.7M cases) is large. We will scope experiments to a representative subset (e.g., federal appellate opinions, ~500K cases) to remain within GPU memory and storage constraints. Fine-tuning Transformer encoders will require GPU access (Google Colab Pro or university HPC cluster).
3. **Data Quality:** OCR artifacts exist in older digitized cases from the CAP dataset. Preprocessing will include text normalization and filtering of cases with low OCR confidence scores. The r/legaladvice Q&A data from Pile of Law is community-generated and may contain inaccurate legal information, which we will use only for intent/question classification, not as ground truth for legal accuracy.
4. **Scope:** This study compares retrieval architectures, not generation models. We hold the generator constant to isolate retrieval effects. Results may not generalize beyond U.S. federal case law to other legal jurisdictions or languages.